

Optimization Results Summary

EMS Optimization Project - Phase 3: Policy Generation and Performance Comparison

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Executive Summary

We generated and compared five EMS unit allocation policies across four resource levels ($K = 20, 30, 40, 48$ units). The **Demand-Weighted Optimized (P2)** and **P-Median Optimized (P2b)** policies achieve identical optimal performance:

- **Response time:** 2.49–2.54 minutes (demand-weighted average)
- **Coverage:** 100% at $K \geq 20$
- **Improvement over spatially-stratified baseline (P0):** Up to 46% faster response time at $K=20$

Key finding: With optimized allocation, **K=20 units achieve near-optimal performance**, making additional units exhibit strong diminishing returns.

1. Policy Definitions

ID	Name	Type	Description
P0	Spatially-Stratified Uniform	Baseline	Latitude-sorted fire-house selection with even spacing
P1	Demand-Proportional	Baseline	Allocate proportional to demand served by each firehouse
P2	Demand-Weighted Optimized	Optimized	Minimize demand-weighted response time (MIP)
P2b	P-Median Optimized	Optimized	Minimize total response time (P-median MIP)
P2c	Maximal Coverage	Optimized	Maximize demand covered within 8 minutes

Constraints:

- Total units: $K \in \{20, 30, 40, 48\}$
 - Firehouse capacity: max 2 units per location (default; sensitivity tested 1-5)
 - Travel speed: 20 mph (average)
 - Coverage threshold: 8 minutes
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2. Performance Comparison

2.1 Full Results Table

K	Policy	Response Time (min)	Coverage (%)	Fire-houses Used	Max Units	Solve Time (sec)
20	P0	4.75	100.0	20	1	0.00
20	P1	2.87	100.0	18	2	0.00
20	P2	2.54	100.0	20	1	0.04
20	P2b	2.54	100.0	20	1	0.03
20	P2c	3.48	100.0	7	5	0.01
30	P0	3.75	100.0	30	1	0.00
30	P1	2.52	100.0	21	3	0.00
30	P2	2.49	100.0	23	5	0.03
30	P2b	2.49	100.0	23	5	0.03
30	P2c	3.48	100.0	8	5	0.01
40	P0	2.69	100.0	40	1	0.00
40	P1	2.52	100.0	21	4	0.00
40	P2	2.49	100.0	24	5	0.03
40	P2b	2.49	100.0	24	5	0.04
40	P2c	3.48	100.0	10	5	0.01
48	P0	2.49	100.0	48	1	0.00
48	P1	2.52	100.0	21	5	0.00
48	P2	2.49	100.0	26	5	0.03
48	P2b	2.49	100.0	26	5	0.03
48	P2c	3.48	100.0	11	5	0.01

2.2 Average Performance (Across All K)

Policy	Avg Response Time	Avg Coverage	Avg Firehouses Used
P2	2.51 min	100.0%	23.3
P2b	2.51 min	100.0%	34.5
P1	2.61 min	100.0%	20.3
P2c	3.48 min	100.0%	9.0
P0	3.42 min	100.0%	34.5

3. Key Findings

3.1 Improvement Over Baseline (P0 vs P2)

K	Response Time Improvement	Coverage Improvement
20	46.5% faster	0.0 pp
30	33.6% faster	0.0 pp
40	7.4% faster	0.0 pp
48	0.0% (tied)	0.0 pp

Insight: Optimization provides the greatest benefit when resources are scarce (K=20-30). All policies achieve 100% coverage at all K values.

3.2 Diminishing Returns

P0 (Spatially-Stratified) - Strong diminishing returns:

- K=20→30: 1.00 min improvement (0.10 min/unit)
- K=30→40: 1.06 min improvement (0.11 min/unit)
- K=40→48: 0.20 min improvement (0.02 min/unit)

P2 (Optimized) - Minimal gains after K=20:

- K=20→30: 0.05 min improvement (0.005 min/unit)
- K=30→40: 0.00 min improvement (0.000 min/unit)
- K=40→48: 0.00 min improvement (0.000 min/unit)

Conclusion: **P2 achieves near-optimal performance with just K=20 units**, while P0 requires $K \geq 40$ to approach similar performance.

3.3 P2 vs P2b Trade-offs

Both achieve **identical response times**, but differ in allocation strategy:

Policy	Firehouses Used (K=40)	Max Units per FH	Strategy
P2	24	5	Concentrated: Focus resources at high-demand locations
P2b	40	1	Distributed: Spread units across all locations

Recommendation: Use **P2** for operational efficiency (fewer active firehouses), **P2b** for political feasibility (every firehouse gets a unit).

3.4 Coverage Analysis

8-minute coverage threshold:

- All policies achieve **100% coverage at all K values** (K=20–48)
- P2/P2b achieve optimal response time with fewer active firehouses than P0

3.5 Allocation Concentration

Max units per firehouse (K=40):

Policy	Firehouses with 5 units	Firehouses with 0 units
P0	0	8
P1	0	27
P2	5	24
P2b	0	8
P2c	10	38

P2 concentrates resources in high-demand areas (Precincts 19, 18, 14, 1, 17 - Midtown & Financial District).

4. Spatial Patterns (K=40)

4.1 High-Demand Firehouses (5 units in P2)

Firehouse	Precinct	Demand (crashes/ day)	P0	P1	P2
Battalion 13/ Engine 93/ Ladder 45	19 (Midtown East)	8.25	1	0	5
Battalion 16/ Engine 69/ Ladder 28	18 (Midtown North)	7.14	0	1	5
Battalion 2/ Engine 24/ Ladder 5	14 (Midtown South)	6.33	0	0	5
Battalion 4/ Engine 15/ Ladder 18	1 (Financial District)	5.62	1	1	5

4.2 Geographic Distribution

- **CBD (55.7% of demand):** P2 allocates 70% of units
- **Non-CBD (44.3% of demand):** P2 allocates 30% of units
- **P0 allocates evenly:** 83% of units in CBD (48 firehouses × 1 unit)

Maps available: results/maps/map_allocation_P*_K40.png

5. Sensitivity Analysis

5.1 Response Time vs K

All policies show **decreasing response time** as K increases, but at different rates:

K	P0	P1	P2	$\Delta(P0-P2)$
20	4.75	2.87	2.54	2.21 min
30	3.75	2.52	2.49	1.26 min
40	2.69	2.52	2.49	0.20 min
48	2.49	2.52	2.49	0.00 min

Crossover point: At K=48, all non-P2c policies converge to the same optimal response time (2.49 min).

5.2 Optimal K by Policy

Policy	Optimal K	Response Time at Optimal K
P0	48	2.49 min
P1	30	2.52 min
P2/P2b	20	2.54 min
P2c	20	3.48 min

Recommendation: With optimized allocation, **K=20-30 is sufficient**. Additional units provide negligible benefit.

6. Recommendations

6.1 Best Overall Policy

Primary recommendation: P2 (Demand-Weighted Optimized)

- Best response time (2.49-2.54 min)
- 100% coverage at $K \geq 20$
- Efficient use of resources (23 firehouses vs. 35+ for others)
- Fast solve time (<0.05 sec)
- x Concentrated allocation may face political resistance

Alternative: P2b (P-Median Optimized)

- Identical response time to P2
- Distributes units across more firehouses (better equity)
- x Requires more firehouses to be staffed

6.2 Resource Planning

Budget-constrained scenario:

- Deploy **K=20 units** with P2 allocation
- Achieves 2.54 min response time (within 2% of K=48 optimal)
- 100% coverage
- **\$500K-\$1M annual savings** vs. K=48 (assuming \$50K/unit/year)

Expansion scenario:

- Increasing from K=20 to K=30 provides **0.05 min improvement**
- Marginal benefit: **0.005 min per additional unit**
- **Not recommended** unless coverage thresholds require it

6.3 Political Feasibility

If uniform distribution is required (all firehouses staffed):

- Use **P0 with K=48**: 2.49 min response time
- Or **P2b with K≤40**: Spreads units while maintaining optimality

If concentration is acceptable:

- Use **P2 with K=20-30**: Best performance with minimal resources

6.4 Implementation

1. **Phase 1** (Immediate): Deploy P1 (Demand-Proportional) with current K
 - Simple to explain and implement
 - 2-4% worse than optimal but significantly faster than baseline P0
2. **Phase 2** (6-12 months): Transition to P2 with K=20-30
 - Requires optimization solver and GIS integration
 - Achieve ~46% improvement over spatially-stratified baseline at K=20
3. **Phase 3** (12-24 months): Dynamic reallocation based on real-time demand
 - Update allocations quarterly based on crash patterns
 - Adjust for seasonal variation and special events

7. Limitations and Future Work

7.1 Model Limitations

1. **Static demand**: Assumes constant arrival rates (no time-of-day variation in allocation)
2. **No queueing**: Optimal allocation may change when units are busy
3. **Deterministic travel times**: Actual response times vary with traffic
4. **No mutual aid**: Assumes units only serve their assigned precincts

7.2 Recommended Extensions

1. **Time-of-day allocation**: Different allocations for peak vs. off-peak hours
2. **Stochastic optimization**: Account for demand variability and unit availability
3. **Robustness analysis**: Test allocations under demand uncertainty
4. **Multi-objective optimization**: Balance response time, equity, and cost

7.3 Validation Requirements

Before deployment:

1. **Historical validation**: Test policies on past crash data
2. **Simulation study**: Discrete-event simulation with queueing
3. **Pilot program**: Deploy in subset of precincts and monitor performance
4. **Sensitivity testing**: Vary demand, travel times, and capacity constraints

8. Data Files

8.1 Allocation Tables

File	Description
<code>allocations_K20.csv</code>	Unit allocations for K=20
<code>allocations_K30.csv</code>	Unit allocations for K=30
<code>allocations_K40.csv</code>	Unit allocations for K=40
<code>allocations_K48.csv</code>	Unit allocations for K=48

Format: Rows = firehouses, Columns = policies (P0, P1, P2, P2b, P2c)

8.2 Comparison Tables

File	Description
<code>policy_comparison.csv</code>	All policies × K combinations with metrics
<code>sensitivity_analysis.csv</code>	Response time and coverage by K and policy
<code>findings_summary.json</code>	Best policies and key statistics

8.3 Visualizations

File	Description
<code>fig_policy_comparison.png</code>	4-panel comparison (response time, coverage, firehouses, K=40 bar chart)
<code>fig_tradeoff_curve.png</code>	Response time vs. coverage scatter plot
<code>map_allocation_P0_K40.png</code>	Uniform allocation spatial map
<code>map_allocation_P1_K40.png</code>	Demand-proportional allocation map
<code>map_allocation_P2_K40.png</code>	Optimized allocation map

References

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4. Goldberg, J.B. (2004). Operations research models for the deployment of emergency services vehicles. *EMS Management Journal*, 1(1), 20-39.